Lecture 24: The Evolution of Shippingport (From the First Commercial PWR to the LWBR)

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Introduction: From Submarine to Grid (and Beyond)

It is important to distinguish the "First PWR" from the "First Commercial PWR."

- **1954 (The First PWR):** The *USS Nautilus* (S2W) launches. This was a military propulsion unit designed for compactness and shock resistance.
- **1957 (The First Commercial PWR):** The **Shippingport Atomic Power Station** goes critical.
- **The Legacy:** Shippingport served as the ultimate testbed for the US nuclear program. Over its 25-year life, the exact same pressure vessel was used to prove the viability of civilian PWRs (Cores 1 & 2) and later to prove the viability of a Light Water Breeder Reactor (Core 3).

The Origins: CVR and Atoms for Peace

Shippingport was not originally designed to power a city; its origins were strictly naval and intensely political.

- **The Cancelled Carrier:** Admiral Rickover's team at Bettis was actively designing the **CVR (Carrier Vessel Reactor)** to propel a new supercarrier. In early 1953, the Eisenhower administration cancelled the CVR project due to budget constraints. Rickover was left with a massive, high-power reactor core design and no ship to put it in.
- **Atoms for Peace:** In December 1953, President Eisenhower delivered his famous "Atoms for Peace" speech at the UN, pledging to use atomic energy for civilian electricity.
- **The Race:** The Soviet Union was already building the Obninsk Nuclear Power Plant. The US Atomic Energy Commission (AEC) needed a commercial reactor online immediately to prove American technological dominance.
- **The Pivot:** Rickover politically outmaneuvered competing civilian designs by pitching his repurposed CVR. The highly enriched naval "Seed" he had already designed would drive a natural uranium "Blanket" to justify it as a civilian testbed.

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1 Part I: The Seed and Blanket (Cores 1 & 2)

Unlike modern commercial reactors that use a nearly uniform enrichment (e.g., 4.5%) throughout, the first Shippingport cores were strictly zoned to bridge the gap between military performance and civilian economics.

- **Core 1 (1957–1964):** Designed to produce 60 MWe (236 MWt).
- **Core 2 (1965–1974):** A major upgrade and power uprate to 150 MWe (505 MWt) utilizing the exact same pressure vessel, proving that significant efficiency gains could be made through core redesign alone.

1.1 The "Seed" (The Naval Driver)

The reactor contained a **Square Annulus (Ring)** of 32 high-performance fuel assemblies.

- **Function:** This region acted as the "spark plug" or neutron driver.
- **Fuel: Highly Enriched Uranium (HEU)**. The fuel was $\approx 93.2\%$ ^{235}U (weapons grade).
- **Geometry: Plate-Type**.
  - *Why Plates?* The Seed generated $\approx 50\%$ of the power in only $\approx 20\%$ of the volume. To prevent melting during transients, the surface-area-to-volume ratio had to be maximized. Thin plates (sandwiched in Zircaloy) provide superior heat transfer.
- **Burnable Poisons:** A fresh HEU core has massive excess reactivity. To prevent having to drive the control rods deep into the core (which causes severe axial flux peaking and uneven burnup), ≈ 170g of **Boron-10** was alloyed directly into the Seed plates. As the ^{235}U burned away, the Boron poison burned away with it, passively flattening the reactivity over time.
- **Control:** Movable control rods in the Seed were made of solid **Hafnium**. Hafnium is unique because its isotopes absorb neutrons to become *other* absorbing isotopes ($^{177}\text{Hf} \rightarrow {}^{178}\text{Hf} \rightarrow {}^{179}\text{Hf}$). The rod maintains its "worth" for decades.

1.2 The "Blanket" (The Civilian Amplifier)

Surrounding the Seed (both inside and outside the ring) were 113 "Blanket" assemblies.

- **Function:** To capture leakage neutrons from the Seed and generate power.
- **Fuel: Natural Uranium** (0.7% ^{235}U).
- **Core 1 Material (Tubes): Uranium Dioxide (UO_2) Pellets in Zircaloy-2 tubes**. Shippingport was the first reactor to use ceramic pellets, proving that oxide fuel could withstand massive irradiation without swelling like metallic uranium. This effectively set the standard for the entire commercial industry.
- **Core 2 Material (Plates):** When the Navy uprated the reactor from 60 MWe to 150 MWe for Core 2, the thermal-hydraulics of the old tubular blanket could not handle the increased heat flux without risking Departure from Nucleate Boiling (DNB). They replaced the tubes with UO_2 wafers compartmentalized in Zircaloy **plates** to drastically increase the heat transfer surface area.

1.3 Reactor Physics: Flux Coupling and The Refueling Strategy

Because the blanket uses natural uranium, its infinite multiplication factor is well below critical ($k_\infty \approx 0.8$). It cannot sustain a chain reaction on its own.

- **Subcritical Multiplication:** The blanket acts as a **subcritical multiplying medium**. The physically thin, highly enriched Seed has massive radial neutron leakage. Fast neutrons leaking out of the Seed enter the Blanket, cause some fast fission in the ^{238}U, thermalize, and then cause thermal fission in the natural ^{235}U.
- **The Isotopic Shift (Uranium to Plutonium):** At the beginning of life, fissions in the blanket are driven by the natural 0.7% ^{235}U. As the core ages, this ^{235}U is severely depleted. However, neutron capture in the ^{238}U breeds fissile ^{239}Pu.
- **Plutonium Equilibrium:** Because ^{239}Pu has a much higher absorption cross-section than ^{238}U, its concentration does not build indefinitely; it reaches a secular equilibrium (roughly 0.5% of the heavy metal mass). By the end of the blanket's life, the original ^{235}U was largely gone, and the majority of the blanket's power was being generated by the *in-situ* fissioning of bred Plutonium!
- **The "Seed Replacement" Strategy:** Because the HEU Seed burned out its reactivity long before the Blanket reached its metallurgical limits, Shippingport did not refuel the whole core at once. During Core 1's lifespan, the reactor was shut down to replace the Seed **three separate times** (Seeds 1, 2, 3, and 4). The exact same natural Uranium blanket remained in place for seven years, effectively transitioning from a Uranium-burner to a Plutonium-burner over its lifetime.

2 Part II: The Light Water Breeder Reactor (Core 3)

In the mid-20th century, a looming "energy cliff" drove nuclear policy. If standard LWRs only burned the 0.7% of natural uranium that is ^{235}U, nuclear energy would be a short-term bridge fuel.

Shippingport Cores 1 & 2 converted some U-238 to Plutonium in the blanket, but they could not breed ($BR < 1.0$). The reactor's final mission (1977–1982) was to prove a breeder could work in a standard PWR lattice.

2.1 Physics: Why Thorium?

In a thermal (water-moderated) spectrum, the number of neutrons produced per absorption (η) for Pu-239 is ≈ 2.07. This leaves only a 0.07 margin to cover leakage and absorption after maintaining the chain reaction and breeding a new atom. To breed U-238, you need Fast Reactors.

To achieve breeding in Shippingport, they switched the fertile material to **Thorium-232**.

$$^{232}\text{Th} + n \rightarrow^{233}\text{Th} \xrightarrow[\text{22 min}]{\beta^-} {}^{233}\text{Pa} \xrightarrow[\text{27 days}]{\beta^-} {}^{233}\text{U} \quad \text{(Fissile)} \tag{1}$$

For U-233 in a thermal spectrum, $\eta \approx 2.30$. This provides a much healthier 0.30 margin of "spare" neutrons to cover losses.

2.2 Reactor Kinetics and The Protactinium Problem

Operating a U-233 core comes with two major physical complications compared to U-235:

- **The Protactinium Parasite:** The intermediate isotope, ^{233}Pa, has a relatively long half-life of 27 days. If the neutron flux in the reactor is too high, the ^{233}Pa will absorb a neutron ($\sigma_a \approx 40$ barns) to become useless ^{234}Pa *before* it has a chance to decay into fissile ^{233}U.
- **The Power Limit:** This parasitic capture destroys the breeding ratio. To minimize it, the core had to operate at a strictly controlled, lower power density. This is why the LWBR (Core 3) was intentionally down-rated back to **236 MWt** (roughly 60 MWe), abandoning the 505 MWt uprate achieved by Core 2.
- **Tighter Kinetics:** The delayed neutron fraction (β) for U-233 is only ≈ 0.0026, compared to ≈ 0.0065 for U-235. This means a U-233 reactor is much closer to prompt criticality during power transients, requiring highly precise, slow-moving reactivity controls.

2.3 The LWBR Design: Movable Fuel

(See Lamarsh, Section 4.5 for diagrams)

Even with the 0.30 margin, the neutron economy was incredibly tight. To utilize it, the LWBR eliminated parasitic control rod poisons entirely.

- **Concept: Geometry Control**. The core was arranged in 12 hexagonal modules.
- **Mechanism:** Rather than inserting poison rods, the entire inner "Seed" assembly of each module was mechanically lifted and lowered along a lead screw mechanism.
- **Leakage Logic:**
  - **Seed Down (Shutdown):** Fuel is axially misaligned. Fast neutrons leak into the surrounding fertile blanket or structural water gap efficiently. $k_{eff} < 1$.
  - **Seed Up (Power):** Fuel aligns perfectly with the blanket. Leakage is minimized. $k_{eff} = 1$.

2.4 The Industrial Challenges of the Thorium Cycle

While the physics worked beautifully, the chemical engineering was far harder than the Uranium cycle.

1. **The "Bootstrap" Problem:** U-233 does not exist in nature. The initial 500 kg loading for the LWBR had to be custom-manufactured at the Savannah River Site by burning massive amounts of U-235 to bombard Thorium targets.
2. **The THOREX Process:** Thorium Dioxide (ThO_2) is incredibly resistant to dissolution. Reprocessing it requires Hydrofluoric Acid (HF), which is extremely corrosive to stainless steel piping, making reprocessing plants vastly more expensive than standard PUREX plants.
3. **The U-232 Gamma Hazard:** Side reactions in the reactor create Uranium-232, which decays into Thallium-208. Thallium-208 emits a highly penetrating **2.6 MeV Gamma Ray**. Unlike fresh U-235, reprocessed U-233 fuel is lethal to handle without heavy shielding (hot cells), drastically increasing fabrication costs.

3 Conclusion: Success and Decommissioning

- **The Success:** The reactor was shut down for the final time in October 1982. Destructive assay of the LWBR core in 1987 proved a final Breeding Ratio of **1.014**.
- **The End of the Thorium Dream:** The discovery of massive, cheap uranium reserves, combined with the difficulty of THOREX processing and the U-232 gamma hazard, kept the Thorium cycle from commercial adoption—even though it could theoretically supply our energy needs for billions of years.
- **A Final First (Decommissioning):** Shippingport was not just the first commercial PWR; it was also the first to be fully decommissioned. Between 1985 and 1989, the site was dismantled, and the intact 900-ton reactor vessel was shipped by barge down the Ohio/Mississippi rivers and through the Panama Canal to the Hanford site in Washington state. The Shippingport site was returned to an unrestricted "greenfield" state.

Rickover's Quote (1977): During the full-power ceremony for the LWBR, Admiral Rickover described the movable fuel assemblies to President Carter:
"So, this whole, huge, 90-ton thing is built to the accuracy of a ***Swiss watch****. That will give you some concept of how difficult, mechanically difficult, in addition to the physics, it is to build a reactor."*

References & Further Reading

1. **Primary Source (Seed/Blanket):** *Shippingport Atomic Power Station: Operating Experience...* WAPD-2965. https://www.google.com/books/edition/Shippingport_Atomic_Power_Station/txcE7JgbMkEC?hl=en
2. **Primary Source (LWBR):** *Water Cooled Breeder Summary Report.* WAPD-TM-1600. https://www.osti.gov/servlets/purl/6957197
3. **Textbook:** Lamarsh, J.R. & Baratta, A.J. *Introduction to Nuclear Engineering.* Section 4.5.
4. **Rickover Quotes:** https://taproot.com/rickover-quotes/

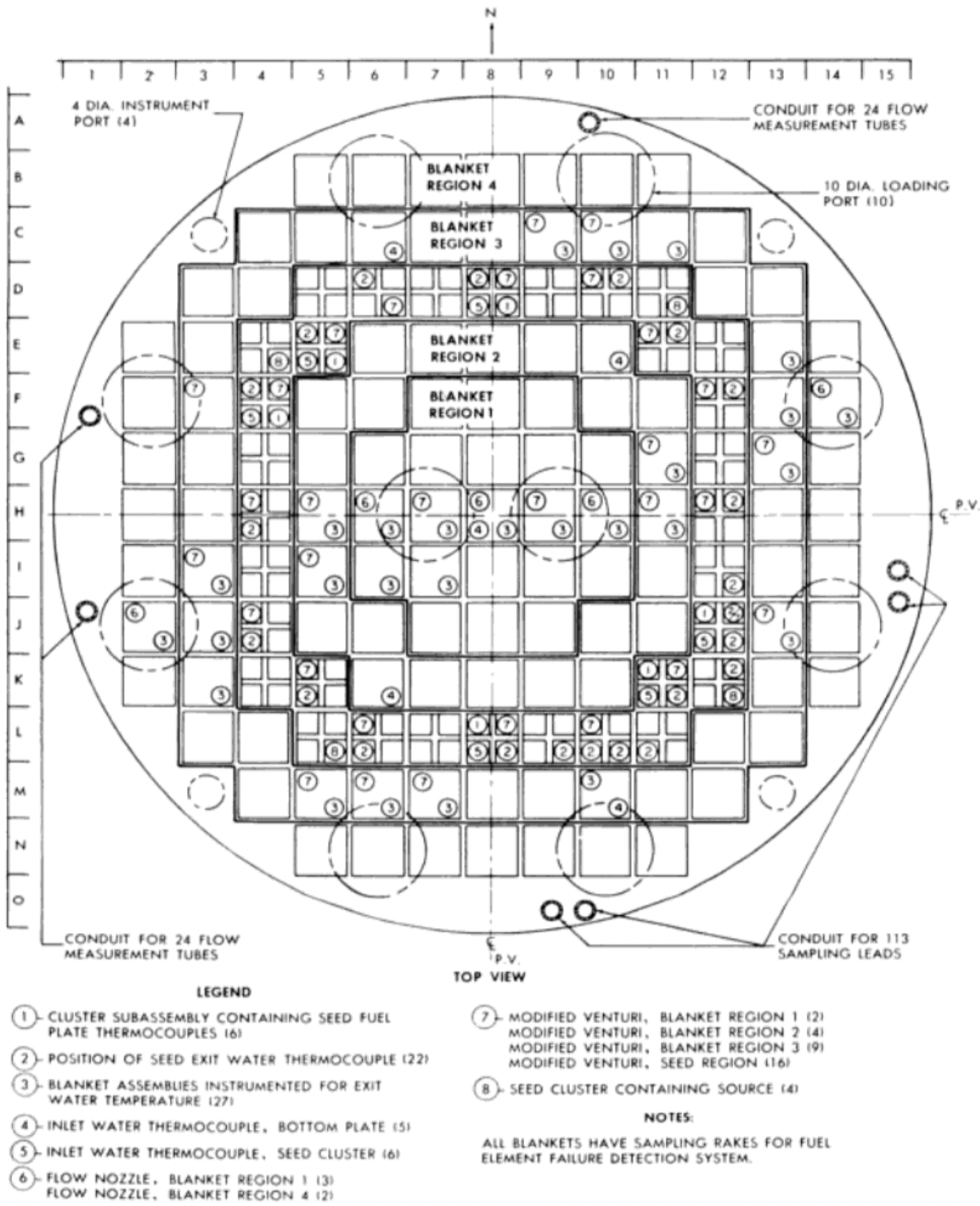

Figure 10. Core 1 Instrumentation - Seed 1.

Figure 1: The Seed-and-Blanket Arrangement. Note the thin "Seed" ring driving the bulk "Blanket."

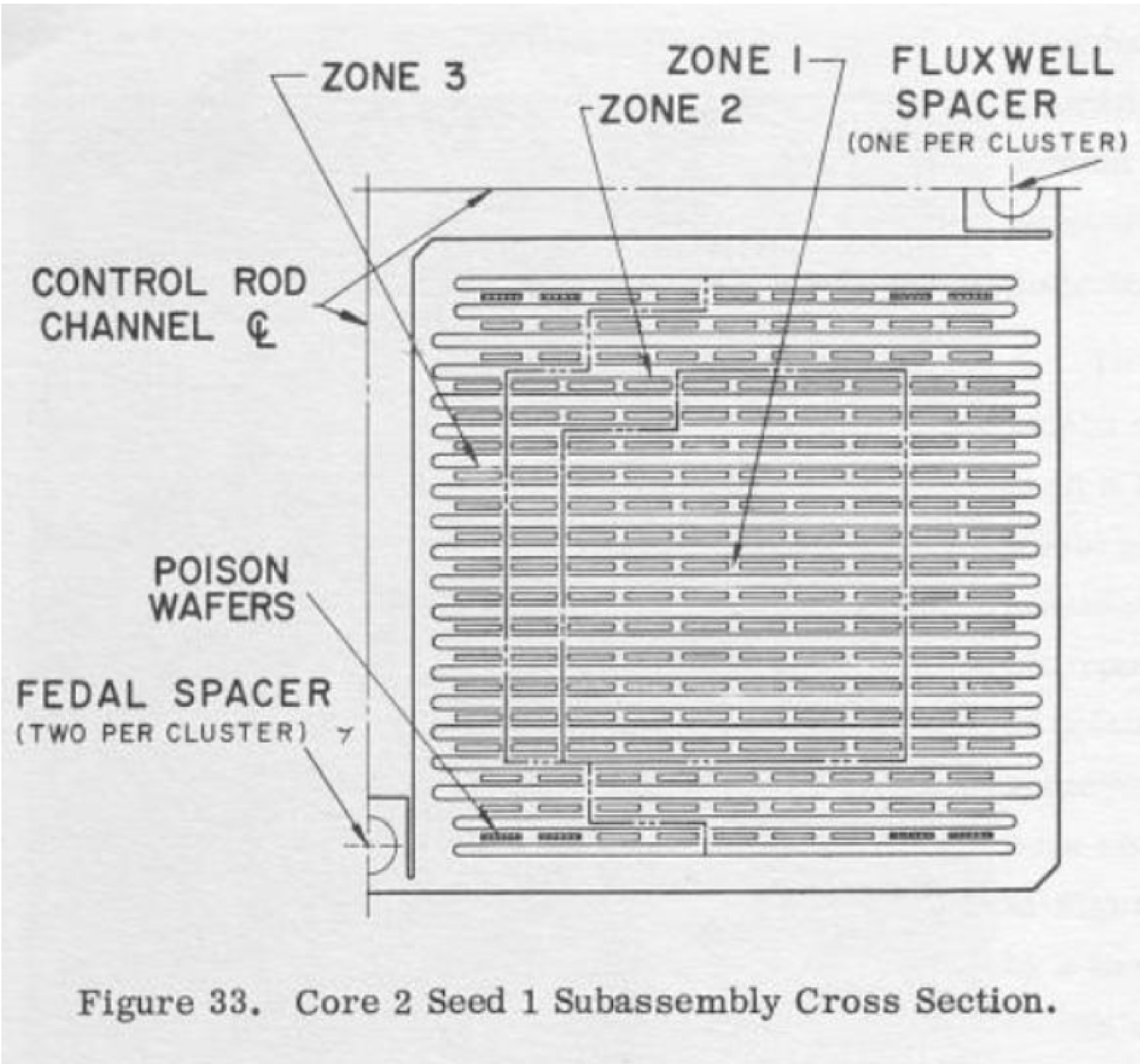

Figure 2: **Detail of the Seed 2 Fuel Cluster.** Quadrants are separated by a central cruciform channel housing the Hafnium Control Rod. *(Reference: WAPD-PWR-2965)*